

Recovering Hidden Fire from RGB: Classical Deformable Models vs. Deep Methods on FLAME-3

CS269 Project — Deformable Models

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1 Problem and Experimental Setup

We study wildfire segmentation on the **FLAME-3** Fire subset (Sycon Marsh aerial captures): 622 spatially aligned RGB/thermal frame pairs at 512×640 . The segmentation target is *where the fire actually is thermally*: the ground-truth (GT) mask is the paired radiometric thermal frame thresholded at $\geq 150^\circ\text{C}$, followed by a 3×3 morphological open/close and a 20-px minimum-blob filter. Thermal is used *only* to define the label; every method receives **RGB only** as its image input.

This framing is the crux of the experiment. In these captures the fire is typically *not* visible as orange flame in RGB—it is obscured by smoke. We measured the red–green opponency ($R-G$, the canonical “fire is red” cue) and found in-fire pixels are no redder than background (mean $R-G \approx 2.5$ in-fire vs. 1.9 background). A trivial color rule therefore *cannot* solve this task by construction; the question is whether a deformable contour or a learned model can recover a largely hidden hot region.

Methods compared. Two classical deformable models—**Kass Snakes** (parametric, single closed contour; `skimage.active_contour`) and **Geodesic Active Contours (GAC)** (morphological level set, multi-component)—are run on the $R-G$ “fire energy” image. Both are evaluated under two initialisations: *oracle* (contour seeded from the GT mask scaled outward $\sim 15\%$, i.e. a known-location prior it only refines) and *color* (seeded from the $R-G$ color floor, no oracle). Three deep baselines take RGB only and are trained against the same thermal masks: **U-Net** (per-pixel CNN, the control), **DALS** (U-Net trunk + a differentiable Chan–Vese level-set evolution), and **Deep Snake** (circular-convolution contour deformation with detector-free coarse-seg seeding). A **color floor** ($R-G$ threshold, no model, no init) is the trivial reference.

Protocol. Classical methods are scored on the 592 frames with non-empty GT (30 empty-GT frames are skipped). Deep methods use a contiguous 70/15/15 train/val/test split (`flame/splits.py`) to avoid temporal leakage between near-duplicate consecutive frames, and are scored on the 94-frame test split. All scores are IoU and Dice on binary masks at native resolution. Deep models were trained 50 epochs each on an RTX 4090 (Adam, $\text{lr} = 10^{-4}$); checkpointed at best validation IoU.

2 Quantitative Results

Table 1 and Figure 1 give the headline comparison.

Method	Init	n	mean IoU	med IoU	mean Dice	med lat (s)
GAC	oracle	592	0.365	0.415	0.499	0.402
Kass	oracle	592	0.198	0.179	0.305	0.277
U-Net	learned (RGB)	94	0.147	0.037	0.207	0.003
DALS	learned (RGB)	94	0.122	0.045	0.182	0.004
Deep Snake	learned (RGB)	94	0.032	0.000	0.032	0.003
Color floor	none	592	0.004	0.000	0.007	0.002
GAC	color	592	0.003	0.000	0.006	0.418
Kass	color	592	0.001	0.000	0.002	0.290

Table 1: Mean/median IoU, mean Dice, and median per-frame latency. Classical: 592 non-empty-GT frames; deep: 94-frame test split.

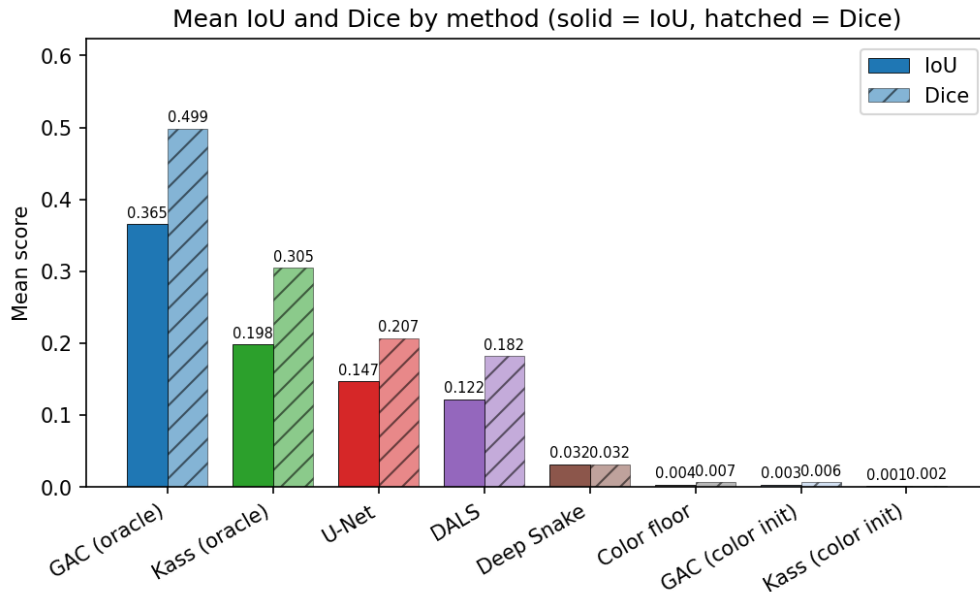


Figure 1: Mean IoU (solid) and Dice (hatched) by method. The oracle contour methods lead; the three RGB-only deep methods sit above the color floor; the no-oracle contour methods collapse onto the floor.

Distributions, not just means. The means hide a crucial difference in *shape*, shown in Figure 2. Oracle-GAC has mean $\approx$ median (0.365 vs. 0.415): it scores moderately on most frames. The deep methods are the opposite—U-Net’s mean (0.147) is far above its median (0.037), and Deep Snake’s median is 0.000. They succeed on a minority of frames (where the fire happens to leave an RGB signature) and score near zero on the rest, producing a heavy right tail rather than uniform competence.

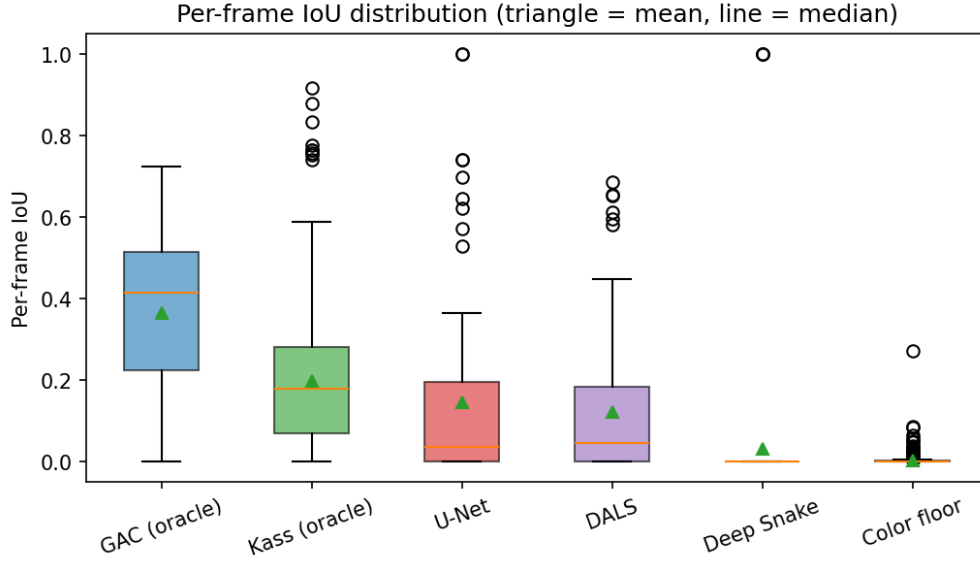


Figure 2: Per-frame IoU distributions. Oracle methods occupy a broad central band; deep methods are concentrated near zero with a high-scoring tail (means, triangles, pulled well above medians).

Robustness to the GT threshold. A reasonable worry is that the 150 °C cutoff is arbitrary and drives the ranking. Re-deriving the GT at 120/150/200 °C (Figure 3, no-oracle init) leaves every no-training method flat and near zero. The color floor and color-init contours are uninformative *regardless* of threshold, so the central conclusion is not a 150 °C artifact.

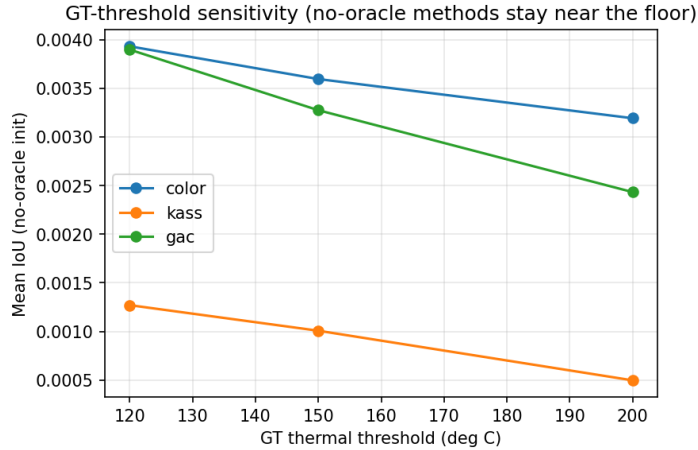
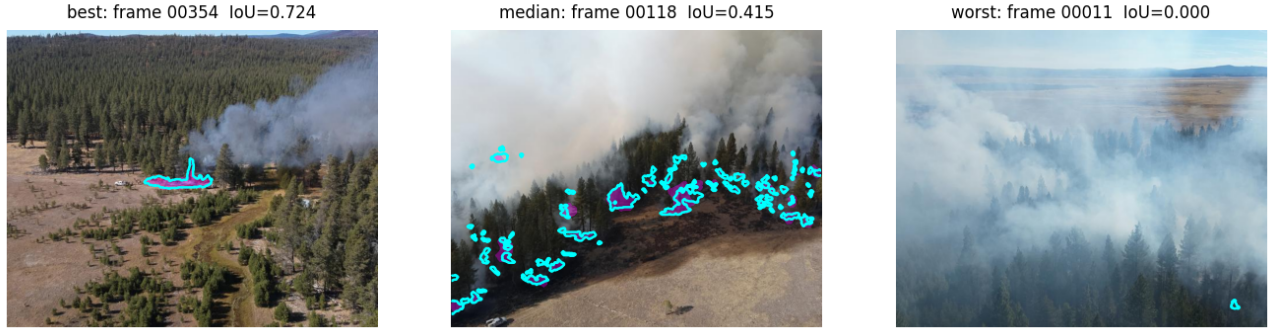


Figure 3: GT-threshold sensitivity (no-oracle init). All no-training methods stay near the floor across 120–200 °C.

3 Qualitative Input/Output Pairs

For each method we show its best, median, and worst test frame by IoU. Cyan is the thermal GT outline; magenta is the method’s prediction. These make the failure modes concrete.

GAC (oracle) (cyan = GT outline, magenta = prediction)



(a) **GAC (oracle)**. Best case tightly recovers a compact front; median over-grows along smoke gradients; worst is a frame where the fire is fully smoke-occluded and the level set finds no boundary to lock onto (IoU 0).

U-Net (cyan = GT outline, magenta = prediction)



(b) **U-Net (RGB-only)**. Best case localises the hot region from RGB alone with no location prior—the key success the contour methods cannot achieve without the oracle. Median/worst predict little or nothing where the RGB carries no usable fire signature.

Figure 4: Qualitative best/median/worst pairs. Full set (Kass, DALs, Deep Snake, color floor) in `report/figs/`.



(a) **Color floor**. Even its “best” frame barely overlaps GT—the $R-G$ cue does not coincide with the thermal fire.

(b) **Deep Snake**. Detector-free coarse-seg seeding is the weak link: when the coarse head misses, no contour is initialised (median IoU 0).

Figure 5: The two extremes: the trivial color floor and the weakest deep method.

4 Conclusions

1. The color floor is ≈ 0 ; the task is genuinely hard. At IoU 0.004 the $R-G$ rule confirms that thermal-defined fire is not recoverable from RGB redness. This is a property of the data (smoke occlusion), not a tuning

failure, and it holds across GT thresholds.

2. Oracle contours win the table but only *refine* a handed-in location. GAC (0.365) and Kass (0.198) lead, but they are initialised from the GT. Their strength is boundary refinement of a known region, not detection.

3. Strip the oracle and the contour methods collapse to the floor (Kass 0.001, GAC 0.003). Seeded from the color prior instead of the GT, they have no usable image gradient to evolve toward and fail completely. *The classical deformable models cannot **find** the fire from RGB—they can only polish a boundary they are already given.*

4. The RGB-only deep methods are the only ones that detect above the floor without a prior. U-Net (0.147) and DALs (0.122) sit an order of magnitude above the color floor and the no-oracle contours, learning a faint RGB signature of the hidden fire. They trail oracle-GAC, but that comparison is apples-to-oranges: the deep methods receive *no* location prior. Their high-mean/low-median profile shows the capability is real but inconsistent—strong on a minority of frames, near-zero on the rest.

5. The contour *head* did not help over a plain CNN. DALs (level-set head) and Deep Snake (contour head) did not beat the plain U-Net; Deep Snake (0.032) underperformed because its detector-free coarse-seg seeding fails whenever the coarse head misses the fire. On this data the bottleneck is *detection from a weak RGB cue*, which a per-pixel CNN handles at least as well as a deformable head bolted on top.

Takeaway. Deformable models are excellent boundary refiners but poor detectors when the image cue is weak; the fair, prior-free comparison is won by the learned per-pixel model. The honest framing of this result is detection from a hidden cue, and under that framing the contour machinery’s value is conditional on a good initialisation it cannot supply itself.

Limitations / next steps. Deep methods saw 50 epochs on a single split; a fair contour-vs-CNN verdict would benefit from cross-validation and a detection front-end for Deep Snake. A natural follow-up is to seed the classical methods from the *U-Net* prediction (a learned, prior-free init) rather than the oracle or the color floor—testing whether contour refinement adds anything on top of a real detector.